DO NOT WRITE IN THIS AREA

12 Show that $(x + 2)(x – 1)(x + 3)$ can be written in the form $ax^3 + bx^2 + cx + d$ where a, b, c and d are integers.

$(x + 2)(x - 1) = x^2 + x - 2$

1 mark
Expand two brackets.

$(x^2 + x - 2)(x + 3)$

$= x^3 + 3x^2 + x^2 + 3x - 2x - 6$

1 mark
Multiply all terms by both x and 3.

$= x^3 + 4x^2 + x - 6$

Therefore $a = 1$, $b = 4$, $c = 1$, $d = -6$.

1 mark
Collect like powers.

(Total for Question 12 is 3 marks)

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